

1. Executive Summary

This project presents an end-to-end securitisation data analytics solution for an auto-loan receivables portfolio. The objective is to transform loan-level and monthly performance data into a structured analytical model capable of supporting portfolio monitoring, credit-risk assessment, investor reporting, waterfall analysis, IFRS 9 expected credit loss assessment, stress testing, and rating-agency reporting.

The solution was developed using Microsoft Power BI with a star-schema data model consisting of loan, borrower, vehicle, geography and date dimensions together with monthly performance, dynamic-loss, static-pool/vintage and delinquency fact tables. The model was designed to provide consistent calculations across portfolio, risk, transaction and reporting views while maintaining clear separation between descriptive dimensions and transactional/performance data.

A comprehensive DAX measure layer was developed covering portfolio balances, loan counts, delinquency, defaults, weighted-average pool statistics, concentration, time-intelligence analysis, vintage performance, waterfall allocation, transaction triggers, IFRS 9 ECL and stress testing. More than 80 measures were documented in a separate DAX Measure Dictionary to improve transparency, reproducibility and model governance.

The Power BI solution contains 10 interactive dashboard pages covering portfolio performance, waterfall and tranche allocation, pool summary, delinquency and early warning indicators, vintage analysis, IFRS 9 ECL, stress testing, investor reporting and rating-agency data-tape analysis. Interactive slicers and analytical visuals allow users to examine portfolio performance by relevant dimensions such as pool, region, origination channel and vintage.

The transaction waterfall was modelled across senior, mezzanine and equity positions. The waterfall allocation was independently validated through an Excel workbook, with the allocation check designed to reconcile total collections against all waterfall distributions. The Power BI waterfall validation currently reports a valid allocation reconciliation, while transaction trigger indicators are monitored separately so that cash-flow reconciliation and trigger status are not conflated.

The IFRS 9 analysis calculates expected credit loss using the $PD \times LGD \times EAD$ framework and provides Stage 1, Stage 2 and Stage 3 views. The model also includes scenario and What-If analysis to evaluate the impact of changes in default assumptions, recovery assumptions, prepayment speeds and interest-rate conditions.

The solution incorporates regional Row-Level Security with separate Central, East, North, South and West roles together with an All-access role. These roles were tested in Power BI to confirm that regional users can be restricted to the appropriate geography while authorised users can access the complete portfolio.

An Excel validation workbook was also prepared containing an ECL crosscheck, waterfall validation and an 80+ measure DAX dictionary. These supporting controls provide an independent audit trail for key calculations and improve the transparency of the analytical solution.

Overall, the project demonstrates how securitisation data can be converted from raw loan-level records into a governed, interactive and risk-focused analytical platform suitable for portfolio monitoring, transaction surveillance, investor reporting and credit-risk analysis.

2. Project Overview and Objectives

2.1 Project Overview

Securitisation involves the pooling of financial assets and the analysis of the resulting receivables portfolio to support funding, risk transfer and investor monitoring. For an auto-loan securitisation transaction, large volumes of loan-level information must be converted into meaningful portfolio and transaction-level indicators.

The purpose of this project was to develop a practical data analytics framework for an auto-loan securitisation portfolio. The framework combines loan characteristics, borrower information, vehicle and collateral information, geography, monthly performance, delinquency history, dynamic loss information and static-pool/vintage information.

The analytical solution was designed in Microsoft Power BI and supported by Excel-based validation. The resulting model provides a single analytical environment for monitoring portfolio performance and credit risk while also producing outputs relevant to investors, transaction stakeholders and rating-agency analysis.

2.2 Project Objectives

The key objectives of the project were:

- 1. Develop a structured securitisation data model**
Build a scalable star-schema model separating dimensions from monthly and transaction-level fact data.
- 2. Measure portfolio performance**
Monitor original balance, current outstanding balance, loan counts, portfolio movements and weighted-average characteristics.
- 3. Monitor delinquency and early-warning indicators**
Analyse current, 30+, 60+ and 90+ day delinquency, default exposure and transaction trigger conditions.
- 4. Perform vintage analysis**
Compare loan cohorts using cumulative defaults, gross losses, recoveries, net losses, prepayments, pool factors and marginal loss rates.
- 5. Implement IFRS 9 ECL analysis**
Calculate expected credit losses using probability of default, loss given default and exposure at default and provide Stage 1, Stage 2 and Stage 3 analysis.
- 6. Model transaction waterfall allocation**
Allocate available collections across senior, mezzanine and equity positions and monitor overcollateralisation, interest coverage, reserve and credit-enhancement metrics.
- 7. Perform stress testing**
Evaluate portfolio and ECL sensitivity under changes in default rates, recovery assumptions, prepayment speeds and interest-rate conditions.
- 8. Automate investor and rating-agency reporting**
Provide pool summary, delinquency, loss and recovery, prepayment, waterfall and trigger information together with rating-agency pool statistics and loan-level data.
- 9. Implement data access controls**
Apply regional Row-Level Security to restrict users according to their assigned geography.

10. Establish independent validation controls

Use Excel-based ECL and waterfall checks and maintain a documented DAX Measure Dictionary for calculation transparency and governance.

2.3 Expected Business Value

The completed solution reduces dependence on manually prepared portfolio reports by centralising calculations and presenting key securitisation indicators through interactive dashboards. It enables users to identify portfolio deterioration, monitor transaction triggers, assess credit losses, compare vintage performance and support recurring investor and rating-agency reporting.

The model also provides a foundation that can be extended with additional reporting periods, pools, transaction structures and risk assumptions without redesigning the complete analytical framework.

3. Data Sources

The project uses four primary datasets covering loan-level characteristics, delinquency history, monthly portfolio performance, and static-pool/vintage performance.

The main auto-loan dataset contains borrower, loan, vehicle, geography, balance, delinquency, credit-risk and ECL fields. The DPD snapshot dataset provides historical delinquency movements and account-level performance. The dynamic monthly dataset provides monthly balances, defaults, losses, recoveries, prepayments and collection metrics. The static-pool dataset supports vintage-level analysis of defaults, losses, recoveries and pool factors.

These datasets were transformed and integrated in Power BI to create a consistent analytical model.

4. Data Preparation and Quality

The source data was reviewed and transformed before modelling. Data types were standardised, dates were converted to appropriate date fields, numerical fields were validated, and analytical bands were created for LTV, DTI, CIBIL, loan size, seasoning, interest rate, original term and delinquency.

Data-quality checks were also incorporated into the reporting model. The rating-agency data tape validation checks Loan ID, Pool ID, Current Balance and EAD completeness. The final validation status was reported as **VALID**.

The model was designed to avoid hard-coded analytical outputs and to calculate key results dynamically from the underlying data.

5. Star Schema Data Model

A star-schema architecture was implemented in Power BI to separate descriptive dimensions from transactional and performance facts.

The main dimensions include:

- **DimLoan** – loan-level attributes and balances.
- **DimBorrower** – borrower credit and financial characteristics.
- **DimVehicle** – vehicle and collateral characteristics.
- **DimGeography** – state and regional information.
- **DimDate** – calendar, month, quarter, year and vintage information.
- **DimDPDBucket** – delinquency classifications.

The main fact tables include:

- **FactMonthlyPerformance** – monthly loan performance and balances.
- **FactDPDSnapshot** – delinquency snapshot history.
- **FactDynamicLoss** – monthly defaults, losses, recoveries and prepayments.
- **FactStaticPool** – static-pool and vintage performance.

This structure improves calculation consistency, filtering performance and scalability.

6. Portfolio Performance Analysis

The portfolio dashboard monitors original balance, current outstanding balance, loan count and changes in portfolio balance over time.

Time-intelligence measures provide MTD, QTD, YTD, month-over-month and year-over-year analysis. Rolling three-month and twelve-month measures are used to identify portfolio trends and abnormal movements.

The dashboard also provides weighted-average portfolio characteristics including WAC, WAM, WALA, LTV, CIBIL and DTI. These metrics provide a concise view of portfolio composition and credit characteristics.

The portfolio trend alert compares current performance with historical rolling statistics and provides an early indication when movements exceed the defined statistical threshold.

7. Delinquency and Early Warning Analysis

The delinquency analysis monitors current, 30+, 60+ and 90+ day delinquency levels. DPD balances and percentages are analysed against the overall portfolio to identify deterioration in payment performance.

The dashboard also provides DPD distribution, monthly delinquency trends, delinquency bands and trigger monitoring. These indicators help identify changes in portfolio credit performance and provide an early-warning view for transaction surveillance.

Transaction trigger measures compare relevant performance indicators with defined thresholds. Trigger status is presented separately from waterfall cash-flow validation so that allocation reconciliation and credit-risk trigger monitoring remain distinct.

8. Waterfall and Tranche Allocation

The waterfall model distributes available collections according to the defined transaction priority of payments. Collections are allocated first to senior fees and senior interest,

followed by senior principal, mezzanine interest, mezzanine principal and the remaining amount to equity.

The model calculates tranche balances, interest obligations, principal allocation, credit enhancement and reserve adequacy. Waterfall validation separately reconciles total collections with the total amount allocated.

Trigger tests for overcollateralisation, interest coverage, delinquency and cumulative losses are monitored independently from the cash-flow allocation validation.

9. Vintage Analysis

Vintage analysis groups loans according to their origination period and tracks portfolio performance by months-on-book.

The analysis includes cumulative defaults, cumulative gross and net losses, recoveries, prepayments, remaining pool balance, pool factor and current delinquency.

Marginal monthly loss rates and cumulative net loss rates are used to identify the timing and development of credit losses across different vintages.

10. IFRS 9 Expected Credit Loss Analysis

The IFRS 9 analysis estimates expected credit losses using Probability of Default (PD), Loss Given Default (LGD) and Exposure at Default (EAD).

The calculated ECL is compared with the provision recorded in the source data, with variance and variance percentage measures used for validation.

ECL is also analysed across IFRS 9 Stages 1, 2 and 3. ECL coverage is calculated by comparing total ECL provision with total EAD.

11. Stress Testing and What-If Analysis

A scenario-based stress-testing framework was developed to evaluate the sensitivity of expected credit losses to changes in portfolio risk assumptions.

The model includes adjustable parameters for default-rate changes, recovery-rate haircuts, prepayment speed changes and interest-rate shocks.

Additional stress scenarios apply progressively higher PD and LGD assumptions to demonstrate the potential impact on ECL and ECL coverage.

12. Investor Reporting

An investor reporting dashboard was developed to provide a consolidated view of transaction performance.

The reporting framework includes Pool Summary, Delinquency Report, Loss and Recovery Report, Prepayment Report, Waterfall Report and Trigger Status.

These sections provide key information on portfolio balances, credit performance, losses, recoveries, prepayments, waterfall allocations and transaction triggers in a format suitable for periodic investor surveillance.

13. Rating Agency Data Tape

A rating-agency-oriented loan-level data tape was developed from the Power BI model to provide key portfolio and loan characteristics required for surveillance and analysis.

The data tape includes Loan ID, Pool ID, origination information, original and current balances, interest rate, remaining term, seasoning, LTV, CIBIL score, DTI, delinquency information, IFRS 9 stage and EAD.

Pool-level statistics include total loans, original balance, current balance, WAC, WAM, WALA, weighted-average LTV, weighted-average CIBIL and weighted-average DTI.

Completeness checks were applied to key fields including Loan ID, Pool ID, Current Balance and EAD. The final data-tape validation status was **VALID**.

14. Row-Level Security and Data Access

Row-Level Security (RLS) was implemented in Power BI to restrict regional portfolio information according to user access.

Separate roles were created for Central, East, North, South and West regions, along with an All-region role for unrestricted portfolio analysis.

The RLS filters are based on the Region field in the geography dimension. Each role was tested using Power BI's role-testing functionality to confirm that the appropriate regional data was displayed.

15. Model Validation and Data Quality

Multiple validation checks were incorporated throughout the project to improve reliability and consistency.

Key validations include loan and pool ID completeness, current-balance completeness, EAD completeness, ECL reconciliation, waterfall allocation reconciliation and rating-agency data-tape validation.

The waterfall check compares total collections with the total allocated amount, while the ECL cross-check compares the calculated $PD \times LGD \times EAD$ value with the reported ECL provision.

These checks provide an audit trail and help identify data, calculation or allocation discrepancies before reporting.

16. Financial Crisis Case Studies

The project also considers historical stress events relevant to securitisation and credit-risk analysis.

The **2008 Global Financial Crisis** demonstrates how deterioration in credit quality, liquidity conditions and funding markets can affect structured-finance transactions and investor confidence. For an auto-loan securitisation framework, the relevant analytical response includes monitoring defaults, delinquencies, recoveries, losses, prepayments and credit enhancement.

The **IL&FS crisis in India** provides a domestic example of how liquidity stress and interconnected financial exposures can affect confidence across financial markets. The project uses this context to emphasise the importance of liquidity monitoring, early-warning indicators, trigger testing, recovery analysis and timely investor reporting.

These case studies are used as analytical context rather than as direct assumptions in the portfolio calculations.

17. Conclusion

This project demonstrates an end-to-end securitisation analytics framework using Power BI, DAX and Excel.

The solution integrates loan-level, delinquency, monthly performance and vintage datasets into a star-schema model and provides portfolio monitoring, delinquency analysis, waterfall allocation, vintage analysis, IFRS 9 ECL, stress testing, investor reporting and rating-agency data-tape capabilities.

The implementation also includes RLS, validation controls and supporting Excel workbooks for independent reconciliation.

Overall, the completed framework provides a structured and repeatable approach for monitoring portfolio performance, credit risk, transaction cash flows and reporting requirements.